User Requirements Specification (URS) for the Music BoomBoxd



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Version History

Version	Date	Author(s)	Changes	State
0.1	03/04/2024	Szymon	Made a beginning on the URS plan.	Finished
1.0	04/04/2024	Szymon	Finished the first version of the URS plan	Finished
1.1	05/04/2024	Szymon	Updated the URS according to small feedback from the WKS teacher	Finished

Glossary

RBAC	Role-Based Access Control
URS	User Requirements Specification
CRUD	Create Read Update Delete
HTTPS	Hypertext Transfer Protocol Secure

Introduction

The User Requirements Specification (URS) outlines the functional and non-functional requirements for the development of the Music BoomBoxd application. This application aims to provide a platform for music enthusiasts to discover new music, connect with like-minded individuals, and engage in meaningful discussions about songs, albums, and artists.

Functional Requirements

FR.1 User Profiles and Interactions

1. The system will allow users to create, view, update, and delete their profiles (CRUD).
2. Users will be able to select their favourite albums and songs to display on their profiles.
3. Users will be able to interact with each other by liking, commenting, and reporting comments on reviews.
4. Users will have the option to receive music recommendations from other users, with mutual consent.

FR.2 Review System

1. Users will be able to write and read reviews of songs, albums, and artists.
2. The system shall provide a form for users to recommend songs, albums, and artists for inclusion in the application.
3. Administrators shall review user recommendations and decide on their inclusion in the application.

FR.3 Content Display and Customization

1. The system shall display the most recent and upcoming albums.
2. Users will be able to set notifications for new album and song releases.
3. The system shall categorise music by year, genre, and popularity for easy search and discovery.
4. Users will be able to customise their music preferences in the settings.

FR.4 User Categorization

1. The users shall be categorized as Professionals, Amateurs or Artists based on their engagement and expertise.
2. Comments and reviews from Professionals and Artists shall be prioritised and displayed prominently.

Non-Functional Requirements

NFR.1 Usability

NFR.1.1 The system will have an intuitive and user-friendly interface to facilitate ease of use of the application.
NFR.1.2 Features such as search filters and sorting options shall be provided for enhanced usability.

NFR.2 Scalability

NFR.2.1 The system architecture will be designed to accommodate future growth and scalability requirements, including increased data volume and user concurrency.
NFR.2.2 Modular and extensible components will be utilised to support incremental updates and enhancements to functionality.

NFR.3 Reliability

NFR.3.1 Comprehensive error handling
NFR.3.2 Multiple data validation check
NFR.3.3 Backup and recovery mechanisms shall be implemented to safeguard against data loss or system failures

NFR.4 Security

NFR.4.1 The system will implement role-based access control (RBAC) to control sensitive information and functionalities based on user roles.
NFR.4.2 The system will implement user authentication mechanisms (e.g., username/password, multi-factor authentication). This will help to verify the identity of users and add another layer of security to the system.
NFR.4.3 The system will implement data encryption protocols to protect confidential data during transmission (e.g., https). This will ensure that confidential data is protected.

NFR.5 Performance

NFR.5.1 The system shall handle concurrent user interactions without significant performance degradation.
NFR.5.2 Response times for common operations should be within acceptable limits, even under peak load conditions.

Use Cases

UC1. User Profile Management

Actor: User
Requirement: FR 1, FR 2
Pre-conditions: Logged in as a registered user.

Main Success Scenario:

1. The User selects “Profile” from the menu.
2. The system displays the user's profile information.
3. The User selects “Delete” to delete their profile.
4. The User selects "Edit" to update their profile.
5. The User makes the desired changes and saves.
6. The system updates the user's profile.
7. The User navigates back to the Dashboard.

Extensions:

3. If the user tries to delete their profile, the system prompts for confirmation.
5. If the user encounters an error while updating their profile, an error message is displayed.

UC2. Interacting with Reviews

Actor: User
Requirement: FR 3
Pre-conditions: Logged in as a registered user.

Main Success Scenario:

1. The User navigates to a review and selects "Like" or "Comment".
2. The system checks if the User’s action violates community guidelines.
3. The system updates the review accordingly.
4. The User can also select “Report” to report inappropriate comments.

Extensions:

2. If the User’s action violates community guidelines, the system displays a warning message.
4. The system prompts for confirmation.

UC3. Writing Reviews

Actor: User
Requirement: FR2
Pre-conditions: Logged in as a registered user.

Main Success Scenario:

1. The User selects "Write Review" for a song, album, or artist.
2. The system displays a form for the user to write their review.
3. The User fills in the review details and submits.
4. The system saves the review and displays it on the relevant page.

Extensions:

4. If the review submission encounters errors, the system prompts the User to correct them.

UC4. Admin Login

Actor: Admin
Requirement: FR1, FR2, FR3, FR4
Pre-conditions: Admin opens App.

Main Success Scenario:

1. The Actor uses his password and username to log in.
2. The system checks if the username and password match the records in the Database.
3. The system displays a Dashboard with three buttons to navigate through the system. Users, Music, and Reviews
4. The actor selects one of the buttons depending on what they want to do.
5. The system opens the appropriate form.
6. The actor does or changes what he wants.
7. The system saves the changes if any were made.

Extensions:

2. The username and password don’t match the records in the Database.
2. The System tells the Actor that the password and username don’t match any known User.

UC5. Music Recommendation

Actor: User
Requirement: FR1
Pre-conditions: Logged in as a registered user.

Main Success Scenario:

1. The Actor uses his password and username to log in.
2. The system checks if the username and password match the records in the Database.
3. The User navigates to the “Recommendations” section.
4. The system suggests music based on the user’s preferences.
5. The User can choose to accept or reject the recommendations.

Extensions:

2. The username and password don’t match the records in the Database.
2. The System tells the Actor that the password and username don’t match any known User.
5. If accepted recommendations, the system updates the user’s profile with the recommended music.
5. If rejected recommendations, the system adjusts its algorithm based on the feedback from the user.

UC6. Notification Settings

Actor: User
Requirement: FR3
Pre-conditions: Logged in as a registered user.

Main Success Scenario:

1. The Actor uses his password and username to log in.
2. The system checks if the username and password match the records in the Database.
3. The User navigates to the “Notifications” section.
4. The system displays options to enable/disable notifications for new releases, comments, likes, etc.
5. The User selects their preferred notification settings.
6. The system saves their settings.

Extensions:

2. The username and password don’t match the records in the Database.
2. The System tells the Actor that the password and username don’t match any known User.
5. If there’s an error in saving settings, the system prompts the User to retry or contact support.